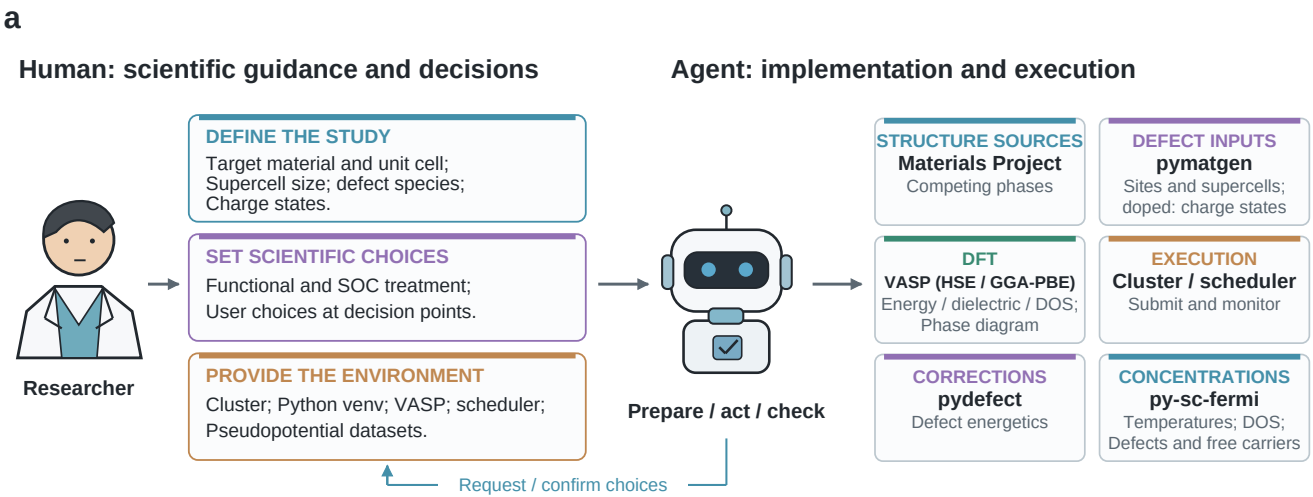

b Full defect calculation and analysis protocol

HOST REFERENCES

1. **Unit-cell structure and electronic structure**
   Relaxation; band gap; DOS
2. **Dielectric constant**
   Electronic and ionic contributions
3. **Phase stability**
   Formation energies; Elemental chemical potentials

DEFECT PREPARATION

4. **Supercell and defects**
   Symmetry-inequivalent sites; charge states; site multiplicities
5. **Validate and execute**
   Check inputs; submit and monitor
6. **Interstitial screening**
   GGA-PBE; select distinct minima

DEFECT CALCULATIONS

7. **Defect relaxation**
   Atomic relaxation; fixed lattice
   Spin polarization; symmetry off
8. **Static calculation**
   Final total energies of defect and defect-free supercells
9. **Validation**
   Inputs; outputs; consistency; provenance

RESULTS AND BACKUP

10. **Postprocessing**
    Charge corrections and formation-energy diagrams
11. **Defect thermodynamics**
    Temperature; charge neutrality; Defect and carrier concentrations
12. **Data backup**
    Preserve hierarchy; verify checksums

◇ **Shared control at every calculation stage**
Validate inputs, check convergence, and resume unfinished relaxations in continue_N directories.

○ User decision / confirmation ◇ Validation / recovery checkpoint